

Genset ECU Field Handbook

Everything known about replacing the engine control unit on a 25 kVA diesel genset — the machine, the 94-pin interface, every circuit with its simulated result, and the two risks that decide whether this ever runs.

Rev 1 13 Sep 2026 6 circuits verified 12 open unknowns Source: 08-GP3-60-001

Sensor Reference →

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01 The machine

The genset is a **Kirloskar KG4-25WS1**. Its engine is a **3R550ETA 4G1** — three cylinders, 1.65 litres, turbocharged and aftercooled, common-rail injection, cooled EGR with a diesel oxidation catalyst and no SCR. It governs at a constant 1500 rpm to make 50 Hz.

That constant speed matters more than it first appears, and it comes up twice in this handbook: once because it makes the timing problem easier than an automotive engine's, and once because the genset's own output frequency is rigidly tied to shaft speed, which turns out to be how a runaway gets stopped.

None of this was given to us. It was derived by taking five hardware constraints from the wiring diagram — three cylinders, common rail, EGR fitted, one catalyst temperature sensor and no urea, a 12 V system — and finding the single engine in Kirloskar's 25–58.5 kVA range that satisfies all five. It is **inferred, not confirmed**; a photograph of the engine rating plate would settle it.

PARAMETER	VALUE	STATUS
Genset	KG4-25WS1, 25 kVA prime, 415 V 3-phase, 50 Hz	Confirmed
Engine	3R550ETA 4G1	Inferred, high confidence
Displacement	1.65 L · 86 × 94 mm · 3 cyl · CR 19.7:1	Confirmed
Rated output	26.5 kW prime @ 1500 rpm	Confirmed
Fuel system	Common rail, metering unit + 3 injectors	Confirmed
Aftertreatment	Cooled EGR + DOC – no SCR, no DEF, no DPF	Confirmed
Emissions tier	CPCB IV+ · GSR 804(E), 3 Nov 2022 · 19–56 kW band	Confirmed
Electrical	12 V, 75 Ah	Confirmed
Panel controller	Kirloskar KG640C	Confirmed
Controller MCU	NXP FS32K148HAT0MLQT, 144-LQFP	Selected

ON THE NAME "GK550"

The engine was first described to us as a GK550. No Kirloskar document uses that string. It is **KG4** — the genset model prefix — run together with **550**, the engine family number inside **3R550ETA**. Use **KG4-25WS1** and **3R550ETA 4G1** when ordering parts; "GK550" will find nothing.

02 The 94-pin interface

The OEM ECU presents one 94-pin connector. The troubleshooting diagram traces **44 of those pins**; the other 50 — power grounds, unused positions, whatever drives the intake throttle — are simply not on a document written for field technicians. A genuine drop-in replacement needs the connector part number to get the rest.

Every wire in the diagram is colour-coded by signal class, and that code is load-bearing rather than cosmetic: it is how we established that the discrete inputs are switched-battery rather than switched-ground, and that the fuel metering unit is driven low-side. The same colours are used below.

- DC positive · sensor supply · injector high side
- DC negative · sensor ground · injector low side
- Analog in · PWM out
- Digital in · frequency
- CAN H
- CAN L

PIN	SIGNAL	CLASS	FRONT-END
21	Battery + feed	● supply	EMI filter → fuse → TVS → reverse-battery FET
04 / 06	V_V_BAT_1R / 2R	● supply	120 k / 10 k divider → ADC · 10 A fused each
09	Synchronization ground	● ?	Function unresolved
11 / 41 / 79 / 34	Boost pressure + air temp	● analog	Combo sensor, 4-pin, shared supply and ground
33	Coolant temperature	● analog	NTC, pull-up divider · ground splices to 34
32 / 35 / 08	Rail pressure	● analog	Ratiometric · dedicated sensor ground
39 / 80	Oil pressure	● analog	Ratiometric · ground splices to 36
15 / 37 / 36	EGR position	● analog	Potentiometer, shares EGR connector
52 / 74 / 30	Crankshaft speed	● frequency	Variable reluctance, differential, twisted pair

PIN	SIGNAL	CLASS	FRONT-END
45 / 46 / 44	Camshaft position	● frequency	Hall effect, already digital
43 / 23	Coolant switch · droop switch	● digital	Switch-to-return + debounce
20 / 24 / 71	Audio abort · override · ignition	● switched batt	Active high — zener clamp, see §03
01 / 02	G_G_B AT1 / AT2	● ground	Switch returns, or main power grounds — unresolved
88	Fuel metering unit	● PWM low side	Low-side driver + current sense
59 / 81	EGR high / low	● PWM	H-bridge, position feedback on pin 37
03	Injector high side — bank A	● high side	Cylinders 1 and 3, joined at splice V5
05	Injector high side — bank B	● high side	Cylinder 2
73 / 07 / 29	Injector low side — cyl 1 / 3 / 2	● low side	Current-controlled peak-and-hold
50 / 69	Main relay · buzzer relay	● low side	N-FET + flyback diode
55 / 77	CAN channel 1	● CAN H ● CAN L	To the KG640C
87 / 86	CAN channel 2	● CAN H ● CAN L	Also wired in the OEM system

INJECTORS SHARE THEIR HIGH SIDES

Pin 03 feeds cylinders 1 *and* 3. Only the low sides are individual. That is five pins, not six, and two boost-fed bank switches rather than three — which matters, because the boost stage is the expensive part of the board. It is safe because on a three-cylinder four-stroke any two cylinders are 240 crank degrees apart, or 26.7 ms at 1500 rpm, against an injection lasting perhaps 3 ms. That separation holds under either candidate firing order, so the unknown firing order stops mattering.

Signals that are not on this connector

The fuel level sender and the water-in-fuel switch both appear in the wiring diagram, and neither reaches the ECU — they route through the XC11/XC1 harness to the panel. They belong to the GCU. Earlier design passes had built front-ends for both.

Also absent from the entire system: any magnetic pickup, any DEF or urea sensor, any LIN bus, any SCR, and any separate overspeed switch.

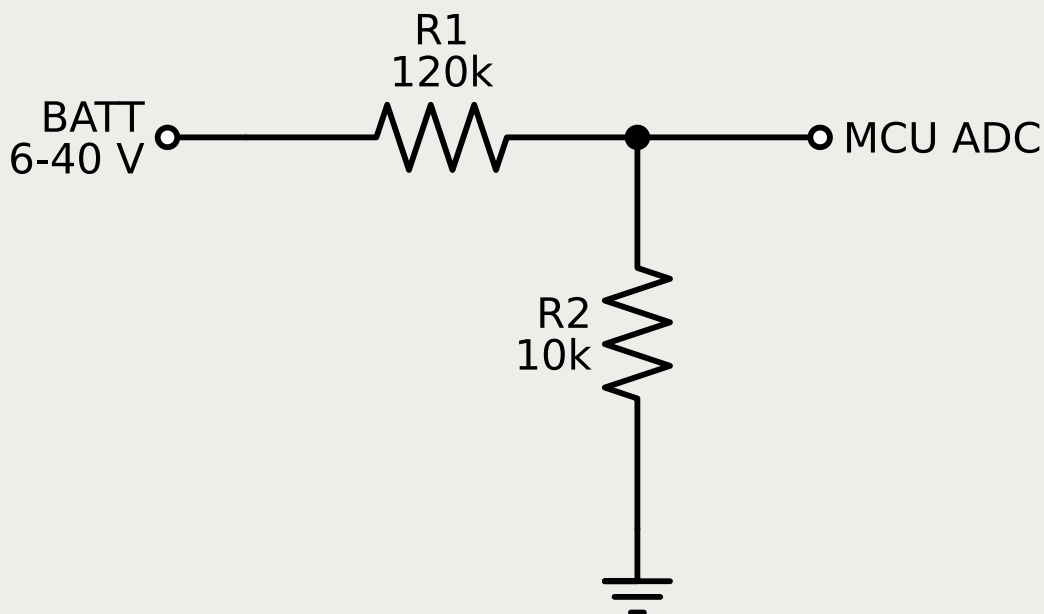
What the diagram labels OVERSIDE SWITCH on pin 24 traces to -S7

Override SS SW — one component, not two.

03 The circuits

Every circuit below exists as a SPICE netlist that runs, with checks that are claims about the circuit rather than about the simulator. A failing check means the design is wrong. All six currently pass — but two of them only after simulation caught errors that had survived four rounds of document review.

Battery voltage sense pins 04, 06

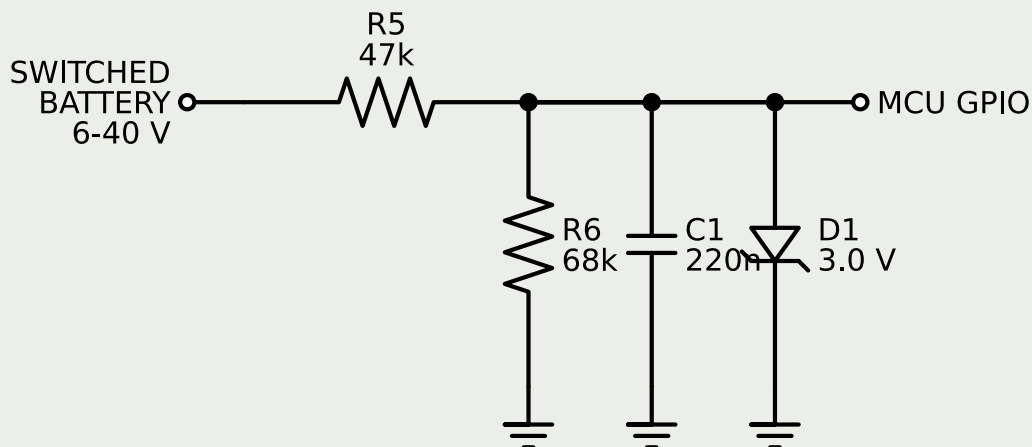


A plain resistive divider scaling the 6–40 V rail into the ADC. Two channels, one per fused rail, which doubles as fuse-integrity monitoring: voltage present downstream means the fuse is intact.

The inherited value was 75 k / 10 k, and it would have destroyed an ADC pin. That ratio was sized for a 5 V ADC; this MCU's analog supply is 3.3 V. At the top of the input range it delivers 4.69 V into a 3.3 V pin. Corrected to 120 k / 10 k.

PASS	Peak output at 40 V in	3.069 V
PASS	Headroom below the 3.3 V rail	0.231 V
PASS	Reading at the 6 V cranking dip	0.462 V · 573 LSB
PASS	Linearity residual	< 1 μ V

Discrete switch input pins 20, 24, 71 · active high



These are switched-battery inputs: the switch connects the pin to battery, not to ground. Earlier documents had them as pull-up-to-5 V with the switch closing to ground — the opposite sense, with a front-end that cannot survive battery voltage.

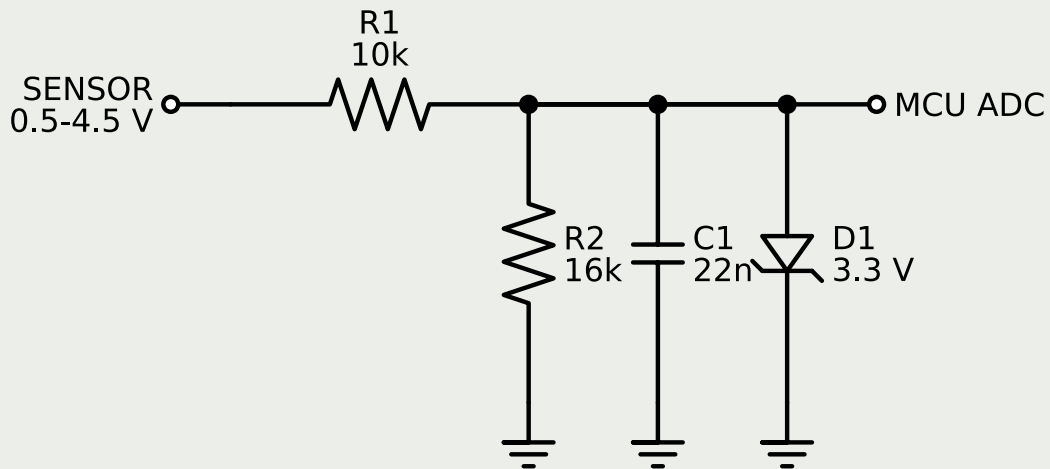
This one cannot be built as a divider at all. Reading logic high at a 6 V cranking dip needs a ratio above $2.31/6 = 0.385$. Staying under 3.3 V at 40 V needs it below $3.3/40 = 0.0825$. Those do not overlap, so no fixed ratio works at both ends — the original was impossible, not merely mis-valued.

A 3.0 V zener sets the clamp level and the divider only biases into it, so the logic level stops tracking the battery and goes flat.

PASS	Level at 6 V cranking dip	2.863 V
PASS	Level at 40 V load dump	2.954 V
PASS	Spread across the whole range	92 mV
PASS	Threshold crossings during a 5-bounce contact	0
PASS	Debounce time constant	6.11 ms

Debounce capacitor went from 100 nF to 220 nF after simulation showed the smaller value let two threshold crossings through a bouncing contact — which firmware would have read as two button presses.

Ratiometric sensor front-end pins 41, 35, 80, 37

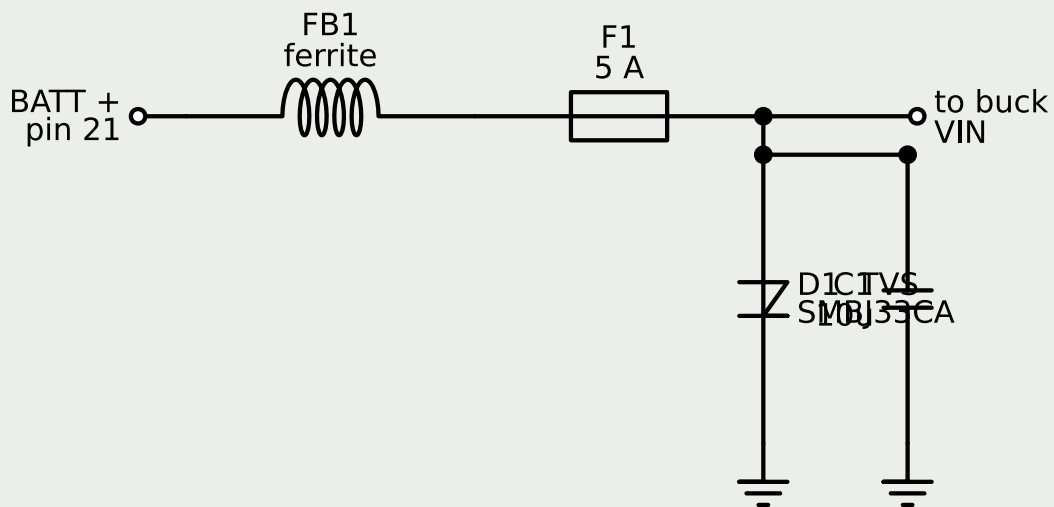


Boost pressure, rail pressure, oil pressure and EGR position are all 0.5–4.5 V ratiometric sensors — their output is a fraction of their own 5 V supply, so the ADC reference has to track that same rail or a drifting supply reads as a pressure change.

Full scale is deliberately placed at 2.77 V, below the 3.3 V clamp rather than hard against the ADC ceiling. That costs about 4% of span and buys a front-end that a 40 V harness short cannot damage.

PASS	Sensor zero (0.5 V) maps to	0.308 V
PASS	Sensor full scale (4.5 V) maps to	2.769 V
PASS	ADC span used	74.6 %
PASS	Low-pass corner	1175 Hz
PASS	Loss at the sensor's own 100 Hz	0.03 dB
PASS	Attenuation at 100 kHz switching noise	38.6 dB
PASS	Pin voltage on a 40 V harness short	3.31 V · 3.7 mA

Transient clamp battery input · ISO 7637-2 pulse 2a

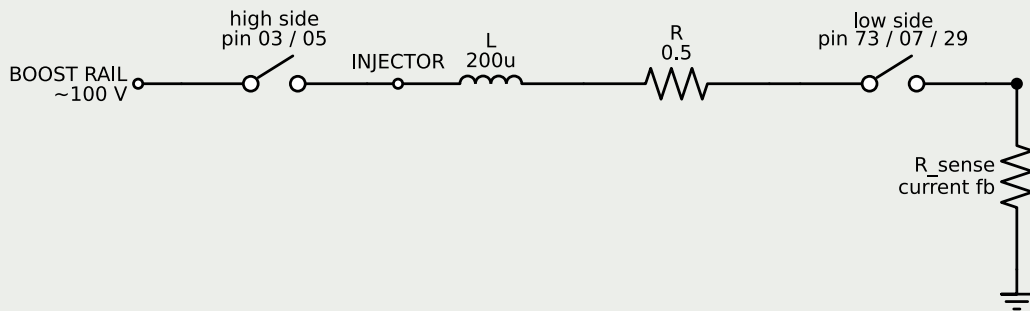


This is the circuit that decides whether the board survives its electrical environment, so it gets simulated rather than assumed. Pulse 2a models a sudden current interruption in another device sharing the harness: **+112 V through 2 Ω for 50 μ s**, at Level IV severity.

PASS	Clamped voltage at the buck input	50.0 V
PASS	Margin inside the LM5164's 100 V rating	2.0 ×
PASS	Recovery after the pulse	12.98 V

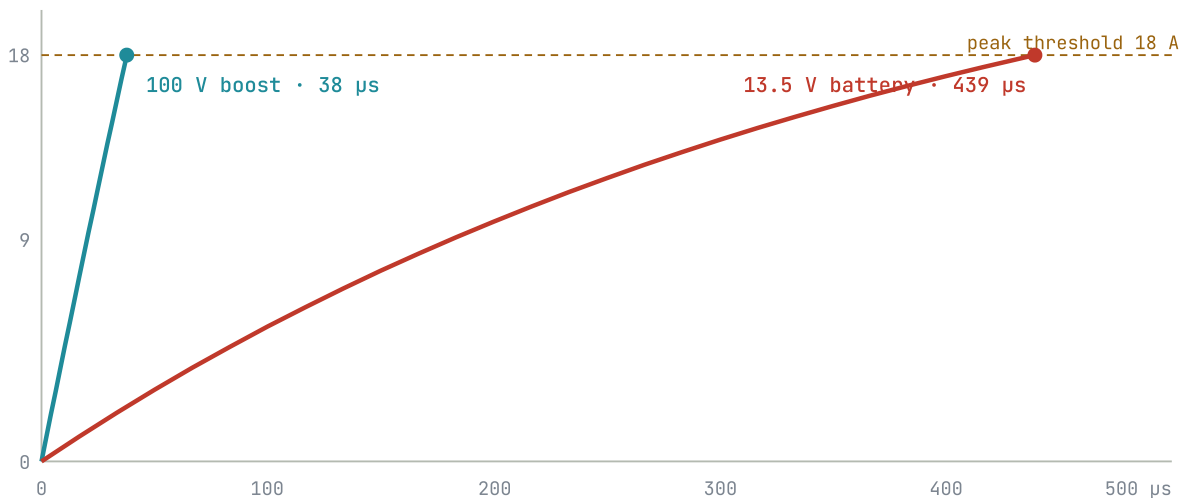
A SPEC TARGET THAT SHOULD BE DROPPED

Earlier documents asked for a clamp "under 42 V". An SMBJ33CA cannot do that — the real part is specified at 53.3 V clamping at rated pulse current, so 50 V here is the component behaving correctly. The 42 V figure was never derivable from a datasheet. Chasing it with a lower-voltage TVS would make the clamp conduct during a legitimate suppressed load dump and force it to absorb that energy. Drop the target; keep the part.



An injector is an inductor with resistance, so current through it cannot change instantly — $di/dt = V/L$. The needle does not lift until current reaches its peak threshold, so every microsecond spent getting there is dead time in series with every injection event.

This is the single most expensive stage on the board. Here is its justification as a measurement rather than an assertion.

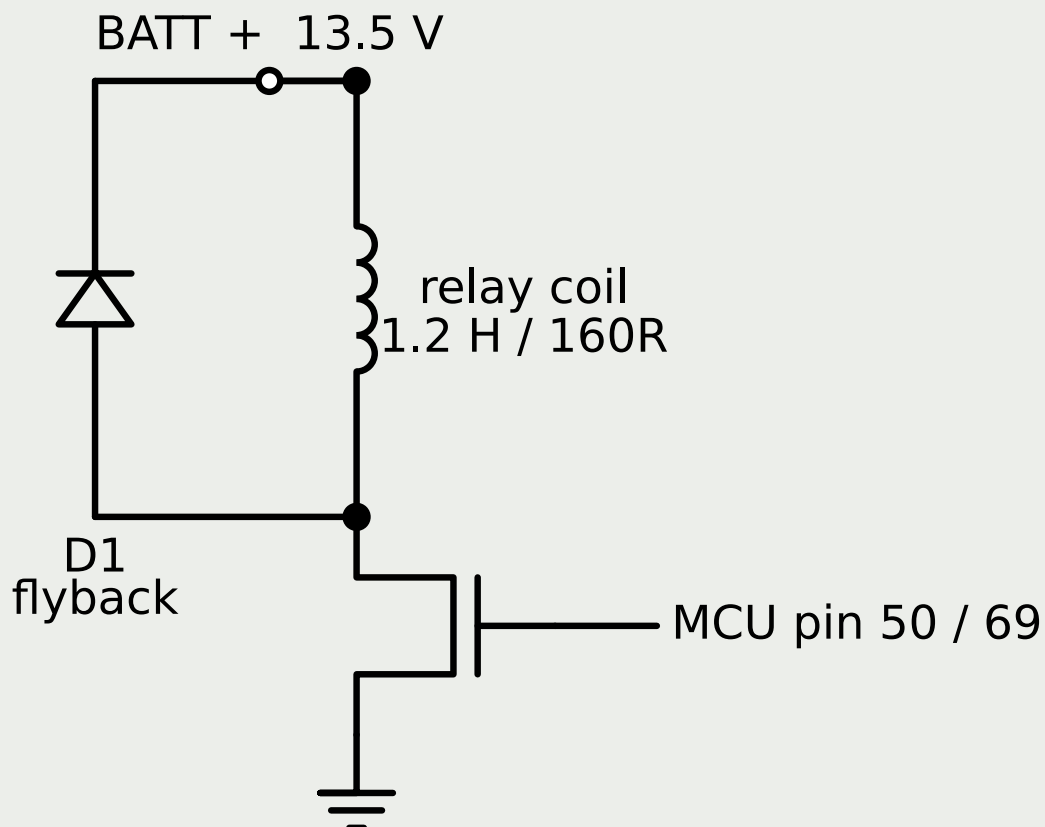


Injector current reaching its 18 A peak threshold. Modelled at 200 μ H / 0.5 Ω , the mid-range of the envelope for solenoid common-rail injectors of this class.

PASS	Time to 18 A from a 100 V boost rail	37.8 μ s
PASS	Time to 18 A from the 13.5 V battery	439.5 μ s
PASS	Boost is faster by	11.6 $\times$
PASS	Boost drive, as a share of a 1 ms injection	3.8 %
PASS	Battery drive, as a share of a 1 ms injection	44.0 %

Without the boost stage, nearly half of the shortest injection event is spent merely lifting the needle. That is why no production common-rail ECU drives injectors from the battery rail, and why the boost converter is not optional.

Relay driver pins 50, 69



A low-side N-FET switching a relay coil. The flyback diode is the entire point: a coil is an inductor, and interrupting its current generates $-L \cdot di/dt$ across it. The diode gives the collapsing field somewhere to circulate.

PASS Drain peak with the flyback diode

14.3 V

PASS Drain peak without it

unbounded

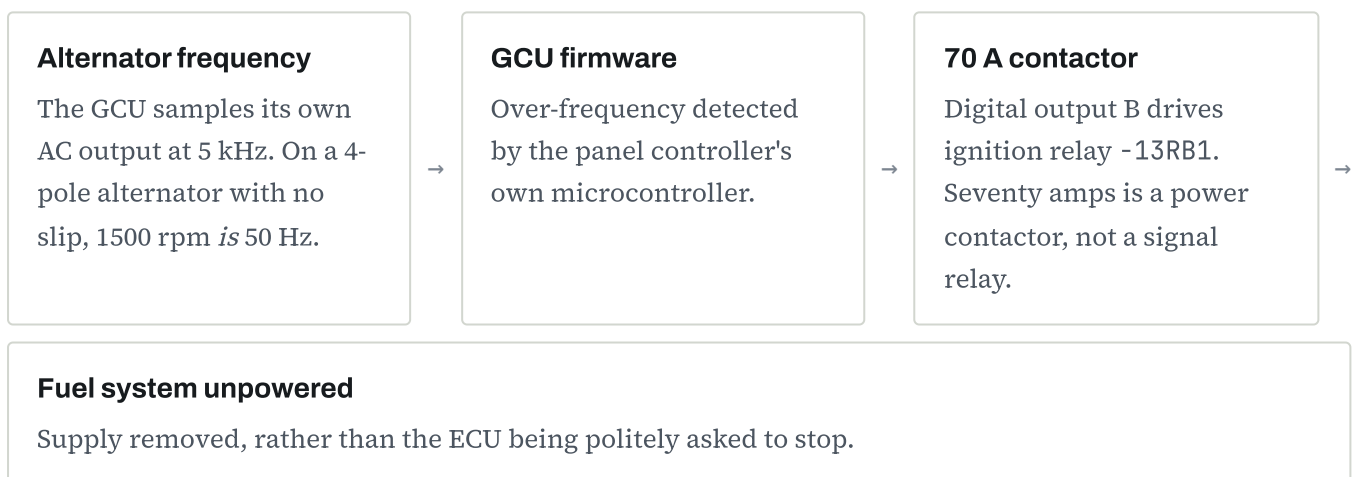
The simulator reports hundreds of kilovolts for the undiode case. Read that as a model artifact – an ideal switch opening instantly across an inductor implies an unbounded voltage. A real MOSFET avalanches at its breakdown rating and turns the coil's stored energy into heat in the die, once per relay operation, until it fails. The magnitude is fictional; the conclusion is not.

04 Stopping a runaway

A 25 kVA set in runaway is a mechanical hazard to anyone near it. The failure that matters is a controller hanging with fuel still flowing — and a controller cannot be its own protection against that.

Our first reading of this system was that engine speed reaches the panel over CAN only, which would mean overspeed protection depended on the very microcontroller and bus most likely to have failed. **That was wrong, and wrong in the pessimistic direction.**

There is no magnetic pickup anywhere — the diagram labels every terminal on the KG640C and none is a speed input, and Kirloskar's manual for the sibling KG640 lists all 42 terminals with no MPU among them. But the panel controller gets engine speed another way entirely, and it can act on it without asking the ECU anything.



Not one link in that chain touches this ECU or the CAN bus. A hung ECU does not defeat it. What remains is a smaller risk: the GCU's own microcontroller is now the single point of failure, and a firmware hang there would defeat the path.

So a standalone magnetic-pickup trip module — a GAC SSW675 or a Murphy HD9063, both with Indian distributors — moves from **mandatory to recommended**. It is a judgement about how much you trust the GCU's firmware, not a missing safety path.

THE GATE THAT REMAINS

All of that rests on one unverified fact: that the ignition relay *removes power* from the fuel system rather than merely signalling the ECU. Confirm it by measurement before any build takes injection authority — de-energise the relay and check for battery voltage at ECU pins 21, 04, 06 and the metering unit's supply pin. **Voltage present means signal-only, and everything in this section reverts.**

05 The two real risks

Neither is a hardware problem, and neither is solved by better circuit design. They are the two things that decide whether this project ends with a working board on a bench or a running genset.

You cannot buy the injector calibration

The drive stage can be designed, built, and fully verified against a known inductive load. That is real engineering progress and it should proceed.

What it cannot do is deliver the *right amount of fuel*. Producing an electrically correct current pulse is not the same as knowing how much fuel that pulse injected — and that mapping, fuel mass against pulse width across the rail pressure range, is proprietary flow-bench data tied to a specific injector's hydraulic geometry. Bosch and Denso ship it with their own ECUs and not otherwise.

It gets harder in the **ballistic region**, where pulses are too short for the needle to reach full lift and delivery becomes steeply non-linear — which is exactly where a genset at light load spends its time. And empirical tuning on a running engine is not a fallback: at 19.7:1 compression, wrong fuelling risks combustion damage and turbocharger overspeed long before trial and error converges.

ROUTE TO THE DATA	COST	LEGALITY
Supplier licence	Structured around OEM production volumes. Realistically out of reach at this scale.	The sanctioned route.
Measure it yourself	Buy injectors, measure delivery directly on a rate-of-injection rig — Bosch tube or Zeuch method. Bounded and controllable with access to a fuel-injection lab.	Entirely legal. First-party measurement of purchased hardware.
Extract the OEM calibration	Cheapest in engineering time if a donor ECU exists, but the binary is undocumented and likely protected.	Contested. Implicates the manufacturer's copyright. Not recommended without legal clearance.

THE ONE WORTH PURSUING

Route two. It is legal, the cost is bounded, and a university with a fuel-injection lab is a far more reachable thing than a Bosch development agreement.

The regulatory position

This is an emissions-certified genset in service. CPCB IV+ type approval attaches to an engine family at the manufacturer level — it is not a per-unit certificate that travels with the machine regardless of what is done to it.

READ THIS BEFORE PLANNING DEPLOYMENT

CPCB's own RECD System & Procedure states that ECU characteristics governing injection timing, air-mass metering and emission-reduction strategy **must not be altered** — and that is under CPCB's own sanctioned retrofit scheme, which separately excludes modifications limited to engine control. General Condition 3 of GSR 804(E) states that no person shall use a genset lacking a valid Type Approval and CoP certificate. No approval pathway for a third-party ECU replacement was found.

This does not touch the engineering. Designing the board, building it, and bench-testing it against simulated crank and cam signals are all unaffected. It bears on putting it into service on a genset that is in use.

And it compounds the calibration problem in a way neither risk shows alone: this engine meets its emissions limits *partly through calibration* — EGR scheduling and injection timing — not through hardware alone. A replacement ECU can therefore move it out of compliance without a single component changing.

This is a decision for the project's owners to take to qualified counsel or to CPCB/ARAI directly. It is recorded here because it is material and had not previously been examined — not to make the decision for anyone.

06 What we don't know

Nothing on this list may be guessed into copper. Most of it needs somebody at the machine with a multimeter and a camera rather than more research — which is worth knowing, because it means one site visit closes most of it.

#	UNKNOWN	BLOCKS	HOW IT CLOSES
U12	Does the ignition relay remove power, or only signal?	The §04 safety gate	Measure with the relay de-energised
U1	The other 50 of 94 connector pins	Whether the board mates at all	Photograph the connector housing
U5	Injector part number	Commissioning, not design	Read the injector body
U4	J1939 PGN set the GCU expects	CAN firmware	Log the live bus with the OEM ECU fitted
U2	Boost air temp — NTC or ratiometric?	One sensor front-end	Measure resistance across the sensor, cold
U6	What pin 09 "synchronization ground" does	Ground plan	Continuity, battery disconnected
U7	Are pins 04/06 sense-only or load feeds?	Power sheet current rating	Measure current at the fuses
U8	Is the 6-pin intake throttle on the ECU?	Whether a throttle driver is in scope	Trace on the engine
U9	Where the catalyst temp sensor goes	Whether an EGT front-end is needed	Trace on the engine
U10	Which signal uses SENT	Possible digital sensor front-end	Inspect the harness
U11	Engine identity confirmation	Nothing — high confidence already	Photograph the rating plate
—	Are pins 01/02 switch returns or power grounds?	Ground plan	Continuity to battery negative

ONE VISIT, MOST OF THE LIST

U12, U2, U6, U7 and the pin 01/02 question are all multimeter measurements. U1, U5 and U11 are photographs. U4 is a CAN log. Done together that is an afternoon, and it unblocks nearly everything currently stalled.

07 Running the simulations

You already have a SPICE engine. KiCad ships `libngspice`, and the harness in the repository drives it directly, so the circuits run and check themselves without installing anything further.

```
.venv/bin/python sim/run_sim.py          # every block
.venv/bin/python sim/run_sim.py injector  # blocks matching "injector"
```

Output is one line per claim, green or red. Because the claims are about the circuit — "reads logic high at a 6 V cranking dip", "never exceeds the MCU's absolute maximum" — a red line means the design is wrong, not that the simulator is unhappy.

The netlists are plain SPICE and are not tied to this harness. They open directly in **KiCad's** built-in simulator (Tools → Simulator), in **LTspice**, or in a standalone **ngspice**. Each carries its own `.control` block, so it runs without further setup.

Read the comments inside each netlist before the netlist itself. They carry the reasoning — why the discrete input is not a divider, why the sensor full scale sits below the clamp, why the 42 V clamp target should be abandoned — which the schematic alone cannot tell you.